

Sharp dyadic mixing-weight stability with a p -concave endpoint

Autonomous proof-discovery packet for arXiv:2501.04656

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Abstract

Figalli, van Hintum, and Tiba ask whether the constant in their sharp Borell–Brascamp–Lieb stability theorem can be made to depend on the mixing weight as $\lambda^{-1/2}$. We prove the conjectured estimate for every dyadic $\lambda = 2^{-m}$ when the endpoint carrying weight $1 - \lambda$ is already p -concave. The mechanism is an exact semigroup identity for p -sup convolution. Homothetic probability densities on cubes show that the power $\lambda^{-1/2}$ is necessary and, additionally, that the constant must blow up at least as $(np + 1)^{-1/2}$ when $p \downarrow -1/n$.

1 Source question and notation

For $a, b \geq 0$, the source uses the zero convention

$$M_{t,p}(a, b) = 0 \quad \text{if } ab = 0,$$

and, when $ab > 0$, the usual weighted power mean

$$M_{t,p}(a, b) = \begin{cases} (ta^p + (1-t)b^p)^{1/p}, & p \neq 0, \\ a^t b^{1-t}, & p = 0. \end{cases}$$

Write

$$S_{t,p}(u, v)(z) := \sup_{z=tx+(1-t)y} M_{t,p}(u(x), v(y)).$$

Theorem 1.6 of Figalli–van Hintum–Tiba [1] proves, for $p > -1/n$, equal-mass f, g , and $h \geq S_{\lambda,p}(f, g)$ with relative integral deficit δ , that f, g are within $O_{n,\lambda,p}(\sqrt{\delta})$ of a common p -concave function, up to a translation. Remark 1.8 asks whether this can be sharpened to

$$O_{n,p}\left(\sqrt{\frac{\delta}{\lambda}}\right), \quad 0 < \lambda \leq \frac{1}{2}.$$

The exact source passage is reproduced in Figure 1.

2 The dyadic endpoint theorem

Theorem 2.1 (Sharp dyadic dependence). *Fix $n \geq 1$ and $p > -1/n$. There is a constant $C_{n,p}$ with the following property. Let $\lambda = 2^{-m}$, $m \geq 1$, and let nonnegative integrable f, g, h on $\mathbb{R}^n$ satisfy*

$$\int f = \int g = 1, \quad h \geq S_{\lambda,p}(f, g), \quad \int h = 1 + \delta.$$

Remark 1.8. As mentioned before, the square root dependence is optimal, as can be seen by considering the one-dimensional example

$$f = \frac{1}{1+\sqrt{\delta}} \mathbf{1}_{[0,1+\sqrt{\delta}]}, \quad g = (1+\sqrt{\delta}) \mathbf{1}_{[0, \frac{1}{1+\sqrt{\delta}}]}, \quad \text{and} \quad h = \mathbf{1}_{\left[0, \frac{1+\sqrt{\delta}+\frac{1}{1+\sqrt{\delta}}}{2}\right]}.$$

Since the family of equality cases for $p = -1/n$ is richer than the class of $(-1/n)$ -concave functions (see Remark [1.5](#)), Theorem [1.6](#) does not hold for $p = -1/n$. In particular, the constant multiplying $\sqrt{\delta}$ must blow up as $p \rightarrow -1/n^+$.

Regarding the dependence on λ , based on corresponding results for the Brunn-Minkowski inequality (see, in particular, [FMP09](#), [BJ17](#), [FMM18](#)), it is natural to expect that the expression $O_{n,\lambda,p}(\sqrt{\delta})$ can be replaced by $O_{n,p}(\sqrt{\delta/\lambda})$. However, proving this bound would require nontrivial work beyond the scope of the current paper, which already introduces several intricate tools and ideas. Therefore, we leave this investigation for future work.

Figure 1: Remark 1.8 on page 5 of arXiv:2501.04656.

If g is p -concave, then there are a p -concave function ℓ and $v \in \mathbb{R}^n$ such that

$$\|f - \ell\|_1 + \|g(\cdot - v) - \ell\|_1 \leq C_{n,p} \sqrt{\frac{\delta}{\lambda}}.$$

The constant is independent of m .

The added hypothesis is one-sided: no concavity is required of f . By swapping the two inputs, the reflected statement holds at weight $1 - 2^{-m}$ when f is p -concave.

3 Exact semigroup identity

Lemma 3.1. If v is p -concave and $r, t \in (0, 1)$, then

$$S_{r,p}(S_{t,p}(u, v), v) = S_{rt,p}(u, v).$$

Proof. Consider a positive term in the nested supremum. It comes from

$$w = tx + (1-t)y_1, \quad z = rw + (1-r)y_2.$$

Put

$$\bar{y} = \frac{r(1-t)y_1 + (1-r)y_2}{1-rt}, \quad \theta = \frac{r(1-t)}{1-rt}.$$

Then $z = rtu + (1-rt)\bar{y}$. For $p \neq 0$, the p th-power coefficients in the nested mean are

$$rtu(x)^p + r(1-t)v(y_1)^p + (1-r)v(y_2)^p.$$

The p -concavity inequality $v(\bar{y}) \geq M_{\theta,p}(v(y_1), v(y_2))$, with the reversal caused by raising to a negative power when $p < 0$, shows that this nested value is at most $M_{rt,p}(u(x), v(\bar{y}))$. The same calculation with logarithms handles $p = 0$. Terms containing a zero input vanish by convention. Taking suprema proves “ $\leq$ ”.

Conversely, in any representation $z = rtu + (1-rt)y$, use $y_1 = y_2 = y$ and $w = tx + (1-t)y$. Monotonicity of the means and the identity

$$M_{r,p}(M_{t,p}(a, b), b) = M_{rt,p}(a, b)$$

give the reverse inequality, including the zero cases. □

4 Proof of the stability theorem

Set

$$u_0 = S_{1/2,p}(f, g), \quad u_j = S_{1/2,p}(u_{j-1}, g) \quad (1 \leq j \leq m-1).$$

Lemma 3.1 gives

$$u_j = S_{2^{-j-1},p}(f, g), \quad u_{m-1} = S_{\lambda,p}(f, g) \leq h.$$

Let $A_j = \int u_j$ and put

$$q = \frac{p}{np+1}.$$

The Borell–Brascamp–Lieb inequality, applied at each midpoint step, yields in the extended sense

$$A_j \geq M_{1/2,q}(A_{j-1}, 1).$$

Backward iteration therefore gives

$$\begin{aligned} A_0^q - 1 &\leq 2^{m-1}(A_{m-1}^q - 1), & q > 0, \\ \log A_0 &\leq 2^{m-1} \log A_{m-1}, & q = 0, \\ 1 - A_0^q &\leq 2^{m-1}(1 - A_{m-1}^q), & q < 0. \end{aligned}$$

Since $A_{m-1} \leq 1 + \delta$ and $2^{m-1} = 1/(2\lambda)$, all three cases imply, whenever δ/λ is bounded by a sufficiently small constant depending only on n, p ,

$$A_0 - 1 \leq C'_{n,p} \frac{\delta}{\lambda}.$$

(For $q < 0$, the same inequalities first force $A_0 < \infty$.)

Now apply Theorem 1.6 of the source only at the fixed weight $1/2$ to the triple (f, g, u_0) . Its deficit is $A_0 - 1$, so it supplies a p -concave ℓ and a translation v with

$$\|f - \ell\|_1 + \|g(\cdot - v) - \ell\|_1 \leq C''_{n,p} \sqrt{A_0 - 1} \leq C_{n,p} \sqrt{\frac{\delta}{\lambda}}.$$

If δ/λ is not small, take $\ell = 0$: the left side is 2 and can be absorbed by enlarging $C_{n,p}$. This proves Theorem 2.1.

5 Sharpness and endpoint blow-up

Proposition 5.1. *Let $Q = [-1/2, 1/2]^n$, $a = 1 + \varepsilon$, and*

$$f_a = a^{-n} \mathbf{1}_{aQ}, \quad g = \mathbf{1}_Q.$$

For $0 < \lambda \leq 1/2$, let $h = S_{\lambda,p}(f_a, g)$ and $\delta = \int h - 1$. Then, as $\varepsilon \downarrow 0$,

$$\delta = \frac{n(np+1)}{2} \lambda(1-\lambda) \varepsilon^2 + O_{n,p,\lambda}(\varepsilon^3),$$

whereas

$$\inf_{\ell, v} (\|f_a - \ell\|_1 + \|g(\cdot - v) - \ell\|_1) = 2(1 - a^{-n}) = 2n\varepsilon + O_n(\varepsilon^2).$$

The infimum is over all p -concave ℓ and all $v \in \mathbb{R}^n$. Consequently, the factor $\lambda^{-1/2}$ is optimal, and any constant in the conjectured estimate satisfies

$$C_{n,p} \geq 2 \sqrt{\frac{2n}{np+1}}$$

in the limit $\lambda \downarrow 0$.

Proof. The sup convolution is constant on $(\lambda a + 1 - \lambda)Q = (1 + \lambda\varepsilon)Q$. Thus, for $p \neq 0$,

$$1 + \delta = (1 + \lambda\varepsilon)^n (\lambda(1 + \varepsilon)^{-np} + 1 - \lambda)^{1/p},$$

while for $p = 0$,

$$1 + \delta = (1 + \lambda\varepsilon)^n (1 + \varepsilon)^{-n\lambda}.$$

Taylor expansion gives the stated quadratic term.

For every ℓ, v , the triangle inequality gives a lower bound by $\|f_a - g(\cdot - v)\|_1$. The overlap of aQ with any translate of Q has volume at most one, with equality when the smaller cube is contained in the larger. Hence this distance is at least $2(1 - a^{-n})$. Equality is attained with $v = 0$ and $\ell = f_a$, which is p -concave under the source's zero convention. Dividing the two asymptotics yields

$$\frac{2(1 - a^{-n})}{\sqrt{\delta/\lambda}} \rightarrow 2\sqrt{\frac{2n}{(np + 1)(1 - \lambda)}}.$$

This proves both conclusions. □

6 Proof intuition, scope, and upgrade audit

Proof intuition. Think of $S_{t,p}(f, g)$ as mixing f with a profile g for time t . Mixing the result with the same p -concave profile again creates two points of g . Concavity lets us collapse those points to their barycenter with no loss, so the two mixing times multiply exactly. A tiny deficit after many half-mixings can then be transported backward to the first midpoint; the source's already-proved midpoint stability finishes the argument.

The unrestricted conjecture is not claimed. Six focused upgrade routes were pursued. Arbitrary λ factors as 2^{-kt} with $t \in (1/4, 1/2]$, but this needs a stability constant known to be uniform on that compact interval, which is not stated in the source. Without endpoint concavity, only $S_{r,p}(S_{t,p}(f, g), g) \geq S_{rt,p}(f, g)$ remains, the wrong direction for backward deficit control. An approximate-concavity bootstrap is circular because sup convolution is not L^1 -Lipschitz. Auditing the source's multistage transport proof did not isolate compact-uniform constants without rebuilding it. Direct transport and level-set routes found no replacement for the missing barycentric collapse. Finally, the cube computation rules out improving either the weight exponent or the endpoint blow-up rate.

Bounded primary-source searches through 11 August 2026 found no later proof of the full conjecture and no prior statement of the dyadic endpoint result. This is a novelty check, not a guarantee of novelty.

References

- [1] A. Figalli, P. van Hintum, and M. Tiba, *Sharp Quantitative Stability for the Prékopa–Leindler and Borell–Brascamp–Lieb Inequalities*, arXiv:2501.04656, 2025.
- [2] H. J. Brascamp and E. H. Lieb, *On extensions of the Brunn–Minkowski and Prékopa–Leindler theorems, including inequalities for log concave functions, and with an application to the diffusion equation*, J. Funct. Anal. **22** (1976), 366–389.